

CSE0611416: Simulation & Modeling Lab

Part A: Course Information

1. **Program:** BSc. in CSE
2. **Course Code:** CSE0611416/CSE413
3. **Course Title:** Simulation & Modeling Lab
4. **Course Type (GED/Core Course/Electives):** Core Course
5. **Year/Level/Semester/Term:** 4th year/ 1st Semester
6. **Academic Session:** Spring 2026
7. **Section:** 7A,7B
8. **Prerequisite Course (If any):** Probability & Statistics; Numerical Methods and Analysis
9. **Credit Value:** 1.5
10. **Contact Hours:** 3.00 hours/Week
11. **Class Schedule:** Tuesday (Room: 411, Period: 1.00 PM – 4.00 PM)
12. **Counseling Schedule:**
 - i. Sunday: 2.00PM-4.00PM
 - ii. Monday: 3.00PM-4.00PM
 - iii. Wednesday: 3:00PM -4:00PM
 - iv. Thursday: 3.00PM-4.00PM
13. **Course Teacher:** Audity Ghosh, Designation: Lecturer, Room: 418(12), Email: audity.ghosh@uits.edu.bd , Cell: +8801755627979, Specialization: Machine Learning, Deep Learning, Computer Vision, Generative Adversarial Networks.
14. **Rationale of the Course:** This course introduces students to the fundamentals of simulation and modeling techniques using Python programming language. The course covers essential concepts including random number generation, statistical hypothesis testing, Monte Carlo simulations, and queueing theory. Students will learn to implement various simulation algorithms, analyze system performance, and apply statistical methods for model validation. The course emphasizes practical applications in engineering systems, providing hands-on experience with Python libraries such as NumPy, SciPy, Matplotlib, and pandas for scientific computing and data visualization.
15. **Course Objectives:**
 1. To gain familiarity with Python and its capabilities for statistical analysis and simulation.
 2. To study probability distribution fitting and its relevance in data analysis and modeling.
 3. To learn various hypothesis testing techniques and their applications in different scenarios.
 4. To explore the implementation of queuing systems, inventory management, and simulation using Python to solve real-world problems.

Part B: Skill Mapping

16. Course Outcome (COs)

No.	COs	BT	CP/W P	CA/E A	KP/WK	Assessment Tools
CO1	Demonstrate foundational knowledge of Python programming, data mining, and statistical methods used in simulation and modeling.	L2	–	–	K2, K3	Q, V

CO2	Implement hypothesis testing techniques and simulation methods to analyze and validate datasets.	L4	-	–	K5	A, LF, CP
CO3	Develop and evaluate simulation and mathematical models for real-world application.	L6	P1, P2, P3	–	K3, K5, K6	PR, R, P

*COs- Course Outcome; BT- Bloom’s Taxonomy; CP/WP- Complex Engineering Problems; EA/WA- Complex Engineering Activities; AT- Assessment Tools; KP/WK- Knowledge Profile; DM&A: Delivery Methods & Activities; LT- Lab Test, LF- Lab Final, Q- Quiz, R- Report, P-Presentation, V- Viva, A- Assignment, CP- Class Performance.

17. Mapping of COs with Program Outcomes (POs)

Course Outcomes (CO)	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	√											
CO2		√										
CO3			√									

Part C: Teaching Learning Approach

18. Course plan specifying contents, COs, co-curricular activities (if any), teaching learning and assessment strategy mapped with COs.

*Commencement of the semester (Autumn 2025): 13 July 2025 and Last class of the semester: 20 November 2025

Week	Topics	Suggested Activity and Teaching Strategy	Assessment Strategy	Corresponding COs
1	Introduction to Python, Basic operations and functions in Python	Students explore Python basics, perform operations, and practice writing functions through examples, diagrams, demonstrations, and collaborative exercises for hands-on learning using Google Colab.	Quiz	CO1
2	Introduction to random number generation in Python. Introduction to testing random number generators, types of dataset discussion.	Students learn random number generation in Python and practice testing random number generators through examples, visual demonstrations, and collaborative exercises for hands-on learning using Python libraries. Data mining concept discussed for understanding dataset.	Quiz	CO1
3	Hypothesis testing concepts in Python, One-sample t-test in Python	Students explore core hypothesis-testing ideas in Python and apply one-sample t-tests using real-life scenario examples, guided demonstrations, and collaborative exercises to analyze results and make informed decisions through hands-on learning with Python tools.	Assignment, Lab Final, Class Performance	CO2
4	Two-sample t-test in Python	Students learned two-sample t-test	Assignment ,	CO2

		concepts in Python by working through real datasets, calculating test statistics, confidence intervals, and p-values, interpreting results to make decisions, and visualizing sample means. They also practiced using NumPy, SciPy, and Matplotlib through guided demonstrations and hands-on examples.	Lab Final, Class Performance	
5	Call for Project Proposal Chi-square goodness-of-fit test in Python	Students learned Chi-square test concepts in Python by working through real datasets, calculating test statistics, expected frequencies, degrees of freedom, and p-values, interpreting results to determine independence or goodness-of-fit, and visualizing observed versus expected counts.	Project	CO3
6	Performing chi-square test in Python	Students were demonstrated with real-life examples included testing whether customer preferences are independent of store location or checking if dice rolls follow a uniform distribution. They practiced using NumPy, SciPy, and Matplotlib through guided demonstrations and hands-on exercises to perform Chi-square calculations, interpret results, and visualize categorical relationships.	Assignment , Lab Final, Class Performance	CO2
7	One-sample Kolmogorov-Smirnov test in Python	Students learned the one-sample Kolmogorov-Smirnov test in Python by comparing a sample's distribution to a theoretical distribution, calculating the KS statistic and p-value, interpreting whether the sample fits the distribution, and visualizing empirical versus theoretical CDFs using SciPy, NumPy, and Matplotlib.	Assignment , Lab Final	CO2
8	Implementation of Monte Carlo simulation in Python	Students learned Monte Carlo simulation in Python by generating large numbers of random samples to model uncertainty and estimate probabilities or expected values. They implemented simulations using NumPy to generate random datasets, performed repeated sampling to approximate outcomes, and visualized results using Matplotlib, gaining hands-on experience in probabilistic modeling and decision-making under uncertainty.	Assignment , Lab Final, Class Performance	CO2
9	Using Python for testing normality. Test for Standard Normal Distribution. Testing for standard normal distribution	Students learned testing for normality and standard normal distribution in Python by analyzing datasets to determine whether they follow a standard normal distribution.	Viva	CO1
10	Discussion of Project Data Collection, Analysis and Report Structure Discussion	Students learned project data collection and analysis by designing methods to gather relevant data, cleaning and organizing datasets, and	Project, Report	CO3

		performing exploratory data analysis to identify patterns and trends. They applied statistical and visualization techniques using Python libraries like NumPy, Pandas, and Matplotlib, interpreted results to draw meaningful conclusions, and documented insights to support decision-making or further research.		
11	Simulation of Single Server Queuing System in Python	Students learned single-server queuing system concepts in Python by modeling the arrival and service of customers in a queue, simulating system behavior, calculating metrics such as average waiting time, queue length, and utilization, and visualizing results. They implemented the system using NumPy and Matplotlib, gaining hands-on experience in discrete-event simulation and performance analysis of queuing systems.	Assignment, Lab Final	CO2
12	Project Demonstration	Students present their simulation and modeling projects. Teaching uses demonstrations, peer review, and Q&A sessions to evaluate understanding, model design, simulation accuracy, and practical implementation skills.	Project, Presentation, Viva	CO1, CO3
13	Lab Final	Students complete a lab final test on simulation and modeling concepts. Teaching uses practical exercises, timed assessments, and interactive review for skill evaluation.	Lab Final	CO2
14	Quiz	Students complete a quiz on simulation and modeling concepts. Teaching uses practical exercises, timed assessments, and interactive review for skill evaluation.	Quiz	CO1

Part D: Assessment Approach

19. Assessment and Evaluation

Assessment components		Marks Distribution
Continuous Assessment (CIA) (100%)	Class Attendance (10) Class Performance (5) Complex Engineering Problem (5)	20%
	Lab Performance	30%
	Quiz (20) Assignment (10)	30%
	Project (5) Presentation (5) Viva (5) Project Report(5)	20%
	Total Marks	100%

20. Grading system

Numerical Grade	Letter Grade	Grade Point
80% and above	A + (A plus)	4.00
75% to less than 80%	A (A regular)	3.75
70% to less than 75%	A – (A minus)	3.50
65% to less than 70%	B + (B plus)	3.25
60% to less than 65%	B (B regular)	3.00
55% to less than 60%	B – (B minus)	2.75
50% to less than 55%	C + (C plus)	2.50
45% to less than 50%	C (C regular)	2.25
40% to less than 45%	D	2.00
Less than 40%	F	0.00

Part E: References

21. Learning Materials

1. **Recommended Readings:** Lecture notes and Necessary documents (Uploaded in Google classroom).
2. **Supplementary Readings**

Text Books:

1. Sweigart, A. (2020). Automate the boring stuff with Python: Practical programming for total beginners (2nd ed.). No Starch Press.
2. Downey, A. B. (2024). Think Python: How to think like a computer scientist (3rd ed.). O'Reilly Media.
3. Rohatgi, V. K., & Saleh, A. K. Md. E. (2015). An introduction to probability and statistics (3rd ed.). Wiley.
4. Ciaburro, G. (2022). Hands-on simulation modeling with Python (2nd ed.). Packt Publishing.

Others (Site):

1. Python Tutorial, Retrieved from: <https://docs.python.org/3/tutorial/>
2. NumPy Documentation, Retrieved from: <https://numpy.org/doc/stable/>
3. SciPy Documentation, Retrieved from: <https://docs.scipy.org/doc/scipy/>
4. Matplotlib Documentation, Retrieved from: <https://matplotlib.org/stable/contents.html>

Part F: Additional Information

- 1) **Program Outcome:** <https://uits.ac.bd/public/cse/>
- 2) **Outcome Based Curriculum:** <https://uits.ac.bd/curriculum-of-cse/>
- 3) **Course Catalog:** <https://uits.ac.bd/undergraduate-programs-cse/>
- 4) **Faculty Information:** <https://uits.ac.bd/faculty-members-cse/>
- 5) **Assessments Rubrics:**
- 6) **Details of Bloom Taxonomy, Knowledge Profile, Complex Engineering Problems and Complex Engineering Activities:**
- 7) **Make-up Exam Procedure:**
- 8) **Attendance Policy:**
- 9) **Academic Calendar:** <https://uits.ac.bd/academic-calendar/>
- 10) **Academic Policies:**
- 11) **Proctorial Rules (Code of Conduct):** <https://uits.ac.bd/wp-content/uploads/2024/01/Code-of-Conduct.pdf>